A Convolutional Neural Network Approach to Predicting Polymer Energy Landscapes from Hi-C Contact Maps

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Abstract: Hi-C contact maps contain information about the three-dimensional structure of a polymer but cannot directly specify the interaction parameters, or energy landscape, needed for simulation. This feasibility study developed a U-Net convolutional neural network (CNN) to infer chromatin interaction potential maps from Hi-C contact maps. The CNN was trained, validated, and tested on non-overlapping windows extracted from six source datasets of human chromatin, and was tuned using one-factor-at-a-time hyperparameter optimization across five model seeds. The model prediction accuracies were compared to an existing graph neural network (GNN) approach developed by Schultz et al. The CNN achieved a lower test loss compared to the GNN but had a longer runtime and greater seed-to-seed runtime variation. It retained performance across genomic distances, captured broad interaction potential patterns, and produced similar contact maps through forward polymer simulation. These findings are preliminary because of the limited number of datasets and model seeds; nevertheless, the U-Net CNN shows promise for inferring energy landscapes from Hi-C contact maps.

1. Introduction

A polymer is a molecule built up of smaller repeating subunits, or monomers. Deoxyribonucleic acid (DNA) and proteins are popular examples of polymers, both of which provide strong evidence that polymers can exist as complex three-dimensional structures: Human DNA stretches to around 2 meters in one-dimension yet fits into a cell nucleus that is micron-sized [1]; and many of a protein's biological functions depend on its specific three-dimensional shape. Evidently, understanding how polymers fold in three dimensions would be important in many fields, ranging from drug discovery to material sciences.

In order to capture how polymers fold in three dimensions, techniques such as Hi-C (High-throughput Chromosome Conformation Capture) were developed. Hi-C maps was introduced by Lieberman-Aiden et al. in 2009 and measures the physical interactions between different polymer loci [2]. The resulting Hi-C maps are matrices that show how different parts of a polymer fold and interact with each other and can be visualised as heatmaps, where the intensity of each matrix entry at position (i, j) represents a population-averaged contact frequency between locus i and locus j.

Although Hi-C maps hold structural information, a polymer cannot be accurately simulated using a raw Hi-C map alone. Consequently, multiple computational works have been done to find the three-dimensional coordinates of polymer segments from Hi-C maps. Deep learning has also been employed to bypass the high computational cost of the corresponding optimisation problems.

However, many coordinate-inferring methods output a single consensus structure which does not fully represent uncertainty in the predicted conformation nor the dynamic nature of a polymer. Furthermore, a coordinate-based representation of a polymer has an arbitrary orientation in a three-dimensional space (i.e. is not rotationally invariant). Brahmachari et al. instead inferred a pairwise interaction energy matrix from the Hi-C map [3]. This approach better captured both structural uncertainty and polymer dynamics. Schultz et al. did the same thing but bypassed the computationally expensive optimisation problem that Brahmachari et al. employed by using an attention-based Graph Neural Network (GNN) to predict the interaction parameters/potentials [4].

While a GNN fits the biological structure of a polymer extremely well, the input of the model is a heatmap image with the output being another heatmap image of the same dimensions. Convolutional Neural Networks (CNNs) are specifically designed and optimised for pattern recognition in image data and could prove to be a faster and more accurate alternative in predicting interaction energy matrices from Hi-C maps.

This summer work had two aims: to develop a CNN from scratch for comparison with an existing GNN approach developed by Schultz et al.; and to determine if a CNN could accurately predict interaction potentials between separated loci, which would allow for the identification of structural features such as chromatin loops.

2. Method

Data

Six datasets were acquired by simulating a (Hi-C) contact map from the respective interaction potential maps of human chromatin. Simulated contact maps were used instead of experimental maps to reduce the uncontrolled experimental variation/noise for this architecture comparison feasibility report. These image datasets were originally (800×800) pixels. Since six datasets were insufficient to train, validate, and test a neural network, (192×192)-pixel non-overlapping windows were extracted from each dataset to increase the number of usable samples (see Fig. 1).

Two models were trained using different window-extraction methods: the first model was trained, validated, and tested on windows extracted along the main diagonal of the datasets to match the symmetric matrix input requirement of the GNN developed by Schultz et al. However, these windows were dominated by interactions between loci in close proximity. Consequently, a second model was trained, validated, and tested using windows extracted from the upper triangle of the datasets. The lower triangle was excluded due to the symmetric nature of the datasets: including the lower triangle would introduce duplicate information into the neural network, worsening computational efficiency and leading to overfitting. The five diagonals nearest the main diagonal were excluded to reduce the influence of proximity-driven interactions, allowing the second model to focus on longer-range interactions that indicate more interesting, informative structures such as chromatin loops. The extracted windows were separated into train, validate, and test sets using a stratified 60:20:20 split. Stratification was done by source map, ensuring that each set contained windows from each dataset. The training procedure/pipeline is depicted in Fig. 1.

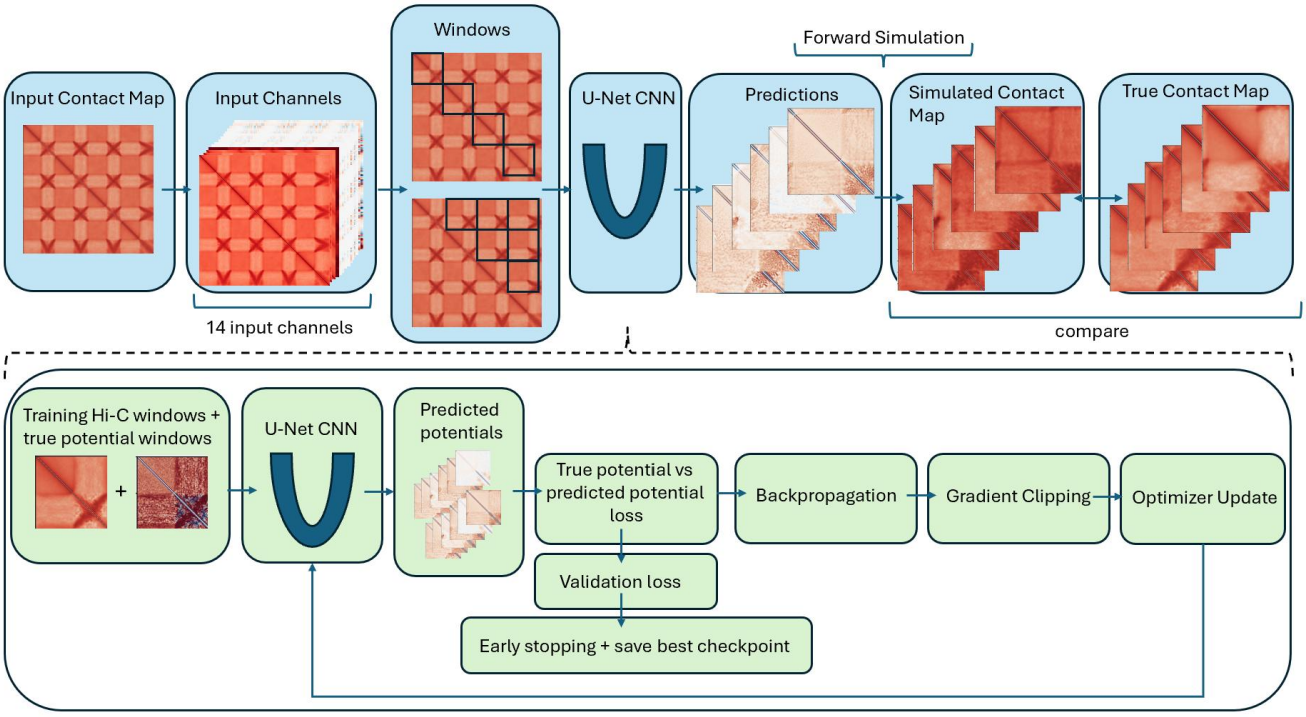


Figure 1. End-to-end U-Net CNN training and evaluation overview. The optimizer updates the learnable model weights/parameters, with the magnitude of each update being determined by the learning rate.

Neural Network

The CNN used a U-Net architecture, allowing the neural network to learn chromatin structures at larger spatial scales, such as compartments (see Fig. 2). Input features were based on those used by Schultz et al., with their node and edge features being adapted into matrix channels suitable for use in a CNN. Likewise, the raw contact map H^{raw} was normalised as done by Schultz et al. by dividing by the mean of the main diagonal (or nearest non-zero diagonal) and then overwriting the main diagonal with 1s:

$$s = \text{mean}\big(\text{diagonal}(H^{raw}, 0)\big),$$
$$H_{ij} = \begin{cases} 1, & i = j, \\ \dfrac{H^{raw}}{s}, & i \neq j. \end{cases} \quad (1)$$

This ensured both models were provided equivalent information, so that any performance differences were due to architectural differences rather than preferable feature selection. The features required adaptation because a CNN operates on matrix inputs in which pixels represent pairwise loci interactions, whereas a GNN processes the loci (node features) and loci interactions (edge features) separately. More specifically, the two edge features used by the GNN were directly used as two input channels to the CNN, encoding:

$$\log\big(H_{ij}\big) \,\&\, \text{mean}(\text{diagonal}(H, |i-j|)). \quad (2)$$

To represent the node features at each pairwise matrix position (i, j), the corresponding ten node features of locus i and locus j were multiplied elementwise. This produced ten input channels that encoded the top ten eigenvectors of a genomic-distance normalised contact map H'_{ij} [4]:

$$H'_{ij} = \frac{H_{ij}}{\text{mean}(\text{diagonal}(H, |i-j|))}. \quad (3)$$

Two additional input channels were included, being the H'_{ij} and the normalised contact map H_{ij}. These extra channels do not introduce additional independent information beyond the GNN node and edge inputs, since this information is inherently encoded in equations 2 and the top ten eigenvectors.

Hyperparameter Tuning

The CNN had 5 tuneable hyperparameters: the learning rate; the model depth; the filter sizes; the base width; and the batch size. Their definitions are described in Fig. 1 and 2.

One-Factor-at-a-Time (OFAT) was the chosen hyperparameter optimisation approach due to time constraints. Each hyperparameter variation was run across

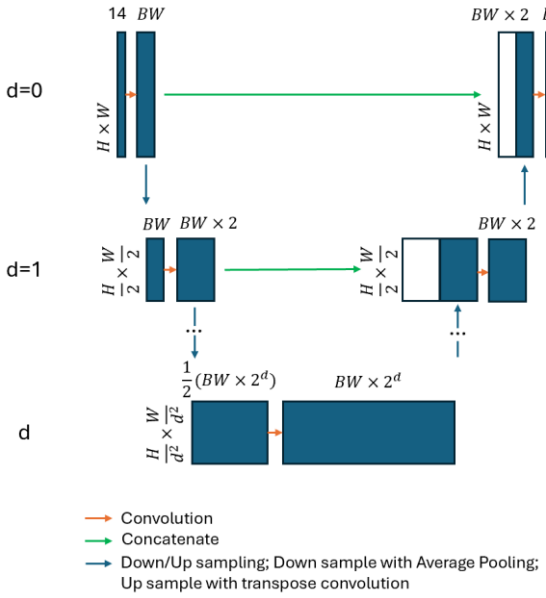


Figure 2. Diagram of the U-Net CNN architecture. H and W are the height and width respectively. BW is the base width, or the starting number of channels after the first convolution, and d is the depth of the model.

five different model seeds to account for stochasticity in initial filter weights and mini-batch ordering. The mean and standard deviation of the validation losses, alongside the train/validation times, were recorded.

Model Evaluation

For a final comparison between the diagonal CNN model and the GNN, the diagonal CNN was run using the tuned hyperparameter settings of the non-diagonal CNN. This was done due to time and resource constraints. Both the diagonal CNN and GNN models were run across 5 different model seeds and were trained and validated on the same dataset splits.

The CNN was evaluated using distance – loss and contact strength – loss analyses; Pearson coefficient and regression-slope analysis of the relationship between predicted and true interaction potentials; and the MAE; MSE; and absolute relative loss. Absolute relative loss is described by:

$$\mathrm{ARL} = \frac{1}{\mathrm{B}}\sum_{b=1}^{B}\frac{\sum_{i,j,k}|y_{b,i,j,k}^{pred} - y_{b,i,j,k}^{true}|}{\sum_{i,j,k}|y_{b,i,j,k}^{true}|} \qquad (4)$$

Finally, the predicted maps from the CNN were used to forward simulate contact maps, which were then compared to the original input contact maps.

3. Results

Table 1. Baseline and tested hyperparameter settings, along with the chosen optimal settings.

Hyperparameter	Baseline	Values tested	Final Selected Value
Learning Rate	1e-4	1e-5, 3e-5, 1e-4, 3e-4, 1e-3, 3e-3, 1e-2	3e-4
Model Depth	3	1, 2, 3, 4, 5, 6	4
Kernel Size	[3, 5, 7]	[3], [3, 5, 7], [3, 5, 7, 9], [3, 5, 7, 9, 11]	[3, 5, 7, 9]
Base Width Size	32	8, 16, 24, 32, 48, 64, 96, 128	64
Batch Size	4	1, 2, 4, 8, 16, 32	4

Hyperparameter values, along with the respective model performance losses and runtimes, can be seen in Fig. 3 and 4. The varied hyperparameter values and the chosen optimal hyperparameter settings are listed in Table 1.

The diagonal CNN model, using the tuned hyperparameter settings, achieved a final absolute relative test loss of 0.3784 ± 0.0917 and a train/validation runtime of 37.43 ± 17.57 min over 5 different seeds. The summary statistics of the optimised diagonal model can be seen in Fig. 5 and Table 3. Weight decay (L2 regularization) was also implemented into a diagonal CNN model as an exploratory extension. The final loss and runtime were 0.350 ± 0.038 and 66.28 ± 11.64 min.

The GNN achieved a final absolute relative test loss of 0.5889 ± 0.0936 and a train/validation runtime of 13.13 ± 5.49 min over 5 different seeds. The final CNN and GNN test loss and runtimes are summarized in Table 2. The non-diagonal CNN achieved a final loss of 0.26 ± 0.16 across 5 model seeds. The simulated Hi-C contact map absolute relative loss across the test set was 0.38 ± 0.17.

Table 2. Mean and standard deviations of the final test losses and runtimes of the CNN and GNN across 5 seeds and across 3 seeds with seeds 44 and 46 excluded. These seeds were excluded as they had resulted in premature stopping in the CNN or GNN due to model instability.

Seed Count	NN	Test Loss	Runtime (min)
5	CNN	0.38 ± 0.09	37.72 ± 17.94
	GNN	0.59 ± 0.09	13.22 ± 5.81
3	CNN	0.34 ± 0.05	45.29 ± 12.75
	GNN	0.55 ± 0.01	17.22 ± 1.44

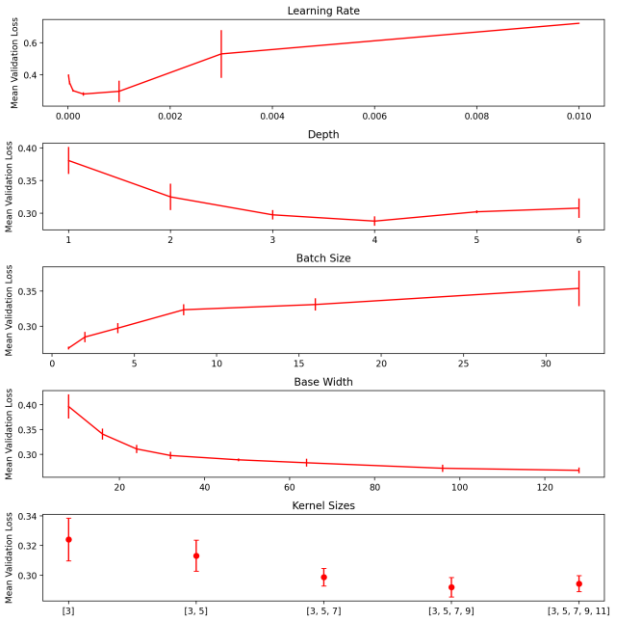


Figure 3. Hyperparameter loss curves following the values in Table 1, with mean and standard deviations over 5 model seeds.

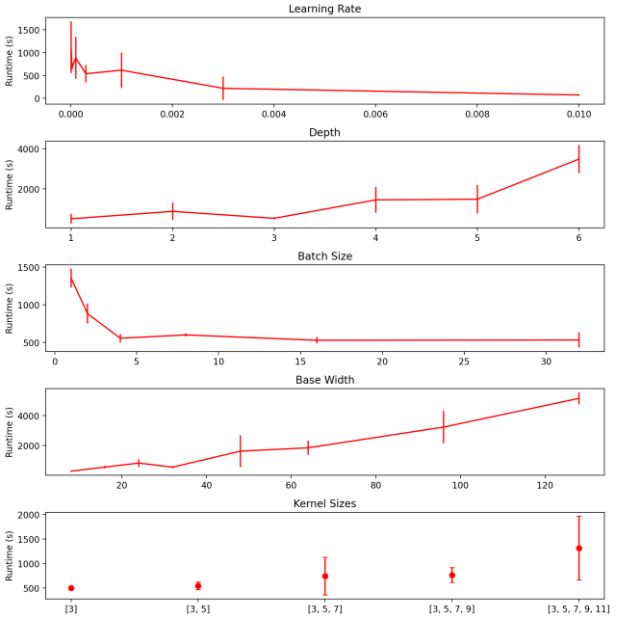


Figure 4. Hyperparameter runtime curves following the values in Table 1, with mean and standard deviations over 5 model seeds.

Table 3. Summary metrics across the test datasets for the diagonal CNN. The absolute relative loss is the loss over each independent matrix element for the model seed, i.e. element-wise absolute relative loss.

Metric	Median	IQR (Q1 – Q3)	Range (min – max)
MAE	0.0650	0.0525-0.0715	0.0473-0.0843
MSE	9.703×10^{-3}	$(7.19 - 11.50) \times 10^{-3}$	$(4.47 - 20.40) \times 10^{-3}$
Absolute relative loss	2.968×10^{4}	$(2.50 - 3.27) \times 10^{4}$	$(2.23 - 3.41) \times 10^{4}$

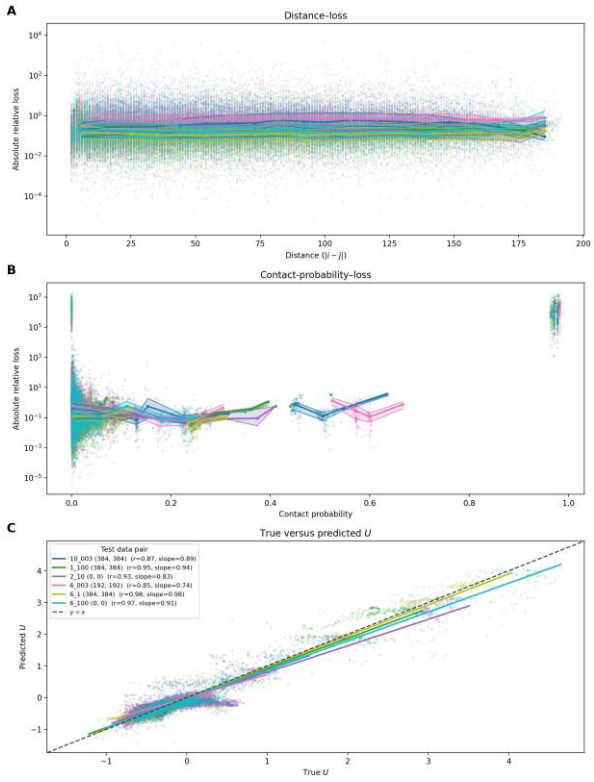


Figure 5. Summary statistics across the six test datasets for the diagonal CNN. Fig 5A. Distance-loss with loss logged. Fig 5B. Contact probability – loss with loss logged. Fig 5C. Pearson coefficients and regression-slopes, with y=x reference line and indication of window origin.

4. Discussion

The L2 regularised diagonal CNN model did not show a clear loss advantage over the original CNN: their mean ± standard deviation intervals overlapped, as did those for runtime. Therefore, neither model can be conclusively determined superior to the other. Despite this, because the original CNN had a lower mean runtime, it was chosen for comparison against the GNN.

Across the same variation of model seeds, the diagonal U-Net CNN model achieved lower test-set loss than the GNN developed by Schultz et al. Although the CNN showed greater seed-to-seed variation, the mean ± standard deviation intervals for both test loss and runtime did not overlap between the models. This indicates that the CNN consistently achieved lower predictive error than the GNN, providing preliminary evidence that the CNN

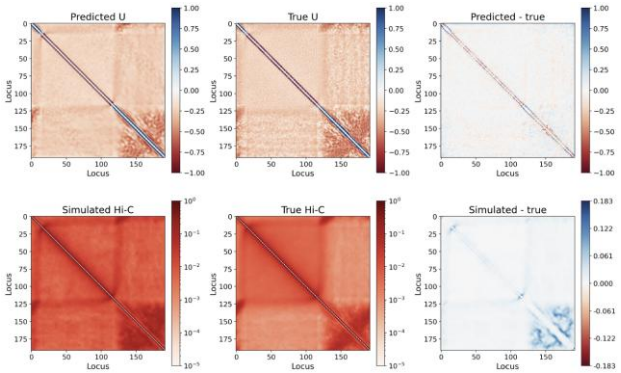


Figure 6. Example of a predicted interaction potential matrix and the simulated Hi-C contact map obtained from it, along with the difference maps for both the interaction potential prediction and the simulated contact map.

architecture can achieve a higher prediction accuracy than the existing GNN approach. However, the small number of seeds limits the statistical strength of this conclusion. The CNN also consistently required longer runtimes than the GNN.

To further interpret the prediction strength of the CNN, Fig. 5 depicts how well the model is predicting interaction potential matrices and implies strong model consistencies. Fig. 5A shows a distance – loss distribution with generally stable median absolute relative losses in each held-out dataset, with the interquartile range (IQR) showing little to no systematic variation as separation distance increased. This indicates that prediction accuracy does not systematically worsen with distance, and that the model performs comparably across most loci separation distances.

Fig. 5B examines the relationship between contact probability and prediction strength. No trend is evident in the median absolute relative loss nor the IQR, indicating that the prediction accuracy is not strongly/systematically dependent on the contact probabilities. The cluster of high losses around contact strengths near one is likely due to the true potential values being near zero around the main diagonal, where the contact probability is one or near one. The larger distribution around contact strengths near zero can be explained by the fact that contact strengths near zero tend to also have true potential values near zero. The reason that true potential values near zero lead to extremely large absolute relative loss values can be seen in Eq. 4.

Fig. 5C analysed the relationship between true potential and predicted potential, showing a median Pearson $r = 0.943$ with a range of 0.85 to 0.98, and a mean slope of 0.881 ± 0.086 with a range of 0.74 to 0.94. The high Pearson r values indicate that the model predicts patterns very well. The high slope values near one indicate that the model predicts accurately but tends to underpredict larger interaction potentials.

Table 3 complements Fig. 5 by providing aggregate metrics across the six held-out datasets. The median MAE was 0.0650 with an IQR of 0.0525-0.0715, while the median MSE was 9.703×10^{-3} with an IQR of $7.189 \times 10^{-3} - 11.5 \times 10^{-3}$. These relatively consistent ranges support the assertion that the

prediction performance was similar across the test set. Conversely, the absolute relative loss's median and IQR are seen to be much larger, which is expected due to the true interaction potentials that are near zero. Consequently, the absolute relative loss metrics should be interpreted alongside the MAE, MSE, and correlation coefficients.

Fig. 6 provides an example of the predicted energy landscape and the corresponding simulated contact map of one held-out dataset. It is evident that the predicted interaction potential map captured the main features of the true interaction potential map, although some local differences remain. When used in a forward polymer simulation, the resulting contact map broadly resembled the true/input contact map. With an absolute relative loss of 0.38 ± 0.17 across the six held-out datasets, there is evidence that the predicted potentials are accurate enough for a forward polymer simulation.

When trained on the non-diagonal windows, the U-Net CNN achieved a final loss of 0.26 ± 0.16. Because the proximity-driven interactions were excluded from the windows, this result provides preliminary evidence that the CNN can accurately infer longer-range loci interactions. However, this loss should not be compared to the GNN as it effectively trained on different dataset distributions. A future direction would be adapting the GNN to accept non-symmetric inputs so that both models could be evaluated on the same longer-range windows.

Altogether, Fig. 5 and 6 and Table 3 provide the preliminary evidence that the U-Net CNN can infer interaction potential landscapes from simulated Hi-C contact maps with sufficient accuracy to then support a forward polymer simulation. Its performance was retained across genomic distances, although the regression-slopes indicated a tendency to underpredict larger interaction potentials. These findings should be interpreted cautiously due to the small number of source maps and model seeds.

There are several major limitations to the generalizability and strength of the CNN approach in this report. Only six source maps were used in the training, validation, and testing of this model. Although non-overlapping windows had increased the number of usable datasets, windows extracted from the same source map are likely to be highly correlated. The stratified split method to distribute the windows between train, validate, and test sets therefore likely allowed for some data leakage, leading to over-optimistic estimates of model performance. The small number of model seeds further limited the statistical inference of the CNN-GNN comparison. Finally, the input source maps were simulated contact maps rather than experimental, meaning that experimental noise was not represented.

Future work should first use a large number of independent, experimental Hi-C contact maps for the training, testing, and validation of the neural network. Further, more model seeds would provide a more reliable statistical analysis on the CNN-GNN comparison. Finally, hyperparameter tuning should be done using a full grid search, or other tuning method, because OFAT does not test for interactions between hyperparameters.

5. Conclusions

In this work, the U-Net CNN achieved lower test losses compared to the GNN across the tested seeds, although it required a longer runtime as well as had a much larger seed-to-seed runtime variation. The model retained its performance over genomic distance, suggesting that the model can support the analysis of longer-range chromatin structures. It was also shown that the model supports forward polymer simulation.

These findings are preliminary due to the limited statistical strength resulting from the small number of model seeds used, as well as the data leakage that likely occurred due to the train, validate, and test splitting methodology. Future work should use independent, full source maps as the input contact maps, vary the model seeds even further, and change the hyperparameter tuning method. Despite this, this feasibility report demonstrated that a U-Net CNN architecture can be a strong alternative to the existing GNN approach developed by Schultz et al.

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